

A Novel Steering System for a Space-Saving 4WS4WD Electric Vehicle: Design, Modeling, and Road Tests

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Abstract—In this paper, we present a steering system for a space-saving four-wheel steering and four-wheel drive (4WS4WD) electric vehicle (EV) with higher maneuverability and flexibility. The proposed system consists of three main parts, namely, an improved two-front-wheel steering (2FWS) mechanism, an omni-directional independent steering (OIS) mechanism integrated with steer-by-wire, and a control strategy for the space-saving steering system of an EV. First, the 2FWS mechanism of the proposed 4WS4WD EV is designed to control the front wheels via the redesigned steering system when the vehicle is traveling at high speeds. Second, a retrofitted OIS mechanism is proposed to achieve an angle range of $-35^\circ \sim +90^\circ$, which is a solid basis for zero radius turning (ZRT) and lateral parking (LP) motion. The driver can control the OIS to turn the four wheels independently, which is assisted by steer-by-wire technologies. Finally, the control strategy for the space-saving steering system of the EV is redefined for the integrated 2FWS and OIS, which can easily handle the EV for high-speed driving or high-maneuverability turning, such as ZRT and LP motion. This system was field tested on a homemade 4WS4WD EV, and the final system simulation and performance evaluation demonstrated the validity of the proposed steering system for the space-saving 4WS4WD EV.

Index Terms—Four-wheel steering and four-wheel drive, two front wheel steering, omni-directional independent steering, zero turning radius, lateral parking, steer-by-wire, control strategy.

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I. INTRODUCTION

ELECTRIC VEHICLES (EVs) have attracted considerable interest as a powerful solution to environmental and energy problems throughout the world [1]. In an EV, the steering system is the most important subsystem that controls the vehicle. With the significant increase in the number of vehicles in urban areas, the ability to turn in narrow areas is urgently needed. However, some gaps still exist in certain aspects of the steering system when a large, sharp turn is required, such as with zero radius turning (ZRT) and lateral parking (LP) motion. With the rapid development of EVs, several innovative technologies have increasingly improved the steering maneuverability, as shown in [2]–[17]. The related literatures include the steering systems of EVs, which include four-wheel-steering (4WS) systems [2]–[6], in-wheel-driving (IWD) systems [7], [8], x-by-wire systems [9]–[11], power steering systems [12], [13], active steering control [14], [15] and human engineering of steering [16].

The combination of the aforementioned technologies has finally been integrated into a four-wheel steering and four-wheel drive (4WS4WD) system to improve the maneuverability and flexibility of EVs [1], [14], [17], [18]. The 4WS4WD EV, as denoted by its name, where each wheel is independently controlled, has many potential applications due to its superior flexibility. The 4WS4WD EV not only takes advantage of the 4WS structure with independents of both steering positions of the front and rear wheels, but also benefits from the four-wheel-driving (4WD) EV, where the individual torque of each wheel can be independently controlled. These approaches, which realize high maneuverability and flexibility of EVs, have been extensively developed.

A. Steering System of an EV

As shown in Fig. 1, the conventional steering system of an EV is confined due to mechanical limits, which have created two inaccessible circular areas at the right and left areas of an EV. A small range of steering angles between -35 degrees and $+35$ degrees decreases the maneuverability in traditional steering configurations [19]. To enhance the mobility of vehicles, the vehicle industry has designed a new type of steering configuration, which is named 4WS for EVs [2]–[6]. The 4WS system of an EV is characterized by the fact that all four wheels



Fig. 1. Conventional steering system of an EV with mechanical linkages.

turn when the driver steers. With the use of rear wheels, the turning radius has been reduced [17], which benefits U-turns, and can be critical to large trucks, tractors and vehicles with trailers. In [20], a Jeep Hurricane, which was designed by the Chrysler Group, was shown to be capable of ZRT by means of complex steering configurations and power transmission mechanisms. Furthermore, another promising technology has been presented with the progression of the steering system for EVs, i.e., the IWD [7], [8]. With the desirable features of being emission-free, space saving and having a large unsprung mass, many in-wheel motors accelerated the prosperity of EVs [21]. Many studies on control systems have also improved the performances of vehicles [8], [22]–[24], which has led to extraordinary potential of vehicle maneuverability and stability.

Upgrading with the propulsion technologies, x-by-wire systems have received considerable attention in the steering system for EVs. Their clear advantages in precision and response speed certainly contribute to control systems, which replace the traditional mechanical control systems and achieve a smoother and more reliable driving experience. In [11], an adaptive control was proposed to realize desirable steering by the steer-by-wire system. Based on a further study of steer-by-wire systems, Wang *et al.* [25] proposed a sliding mode control scheme by building a second-order system for steering. This study contributed to a better understanding of the interactions between components in a steer-by-wire system using detailed modeling. In [13], Naranjo presented a power steering system that can automatically move the steering wheel of a mass-produced vehicle. The experimental results show that the combination of GPS and artificial intelligence-based techniques has performed very well.

B. Omni-Directional Independent Steering of EVs

As integrated components, omni-directional independent steering (OIS) systems respond to the universal challenge of omni-directional vehicle motions. Compared with the existing specially designed wheels for OIS vehicles, such as universal wheels, Mecanum wheels and ball wheels [26], OIS systems with conventional wheels have larger capacities and a greater tolerance for ground irregularities. In [27], Theo *et al.* proposed a wheel module in the context of the control problem that can be used for a large variety of omni-directional mobile platforms for applications in the industry, public and household environments. The authors in [28] presented an innovative omni-directional robot called AZIMUT based on caster wheels. Using a wheel kinematical model, they demonstrated how to compute the mobility and to adjust the steerability of the robot. They proposed a new design paradigm based on the

use of four serial elastic actuators to motorize the steering. To achieve smooth position and orientation tracking of the omni-directional mobile platform based on four caster wheels, a differential flatness-based, point-to-point trajectory planning and control scheme was proposed in [29]. A new gearing mechanism was proposed by Ueno *et al.* in [30]. The method is effective because it is possible to overcome differences in level and can absorb vibrations using normal tires. Wada *et al.* proposed an active caster drive system for omni-directional mobile robots [31]. The proposed mechanism includes a dual-ball transmission to transmit traction power to rotate and orient a drive wheel while distributing velocity components to the wheels and steering axes in an appropriate ratio. The sensor that measures the wheel orientation and the calculations for velocity decomposition are completely removed from a conventional control system due to the determined velocity ratio. This approach simplifies the entire omni-directional control system and is easily controllable. Vehicle industries, research institutions and universities have boldly attempted to improve the maneuverability of EVs based on robotic technologies, such as Pivo2 [32], Hiriko [33], [34] and OK-1 [35].

In the open-access resources, very little research has been published on the control of OIS EVs. In [36], Lam *et al.* presented an omni-direction steer-by-wire system for a 4WIS vehicle with an extended steering interface and a behavior-based steering controller, which improved vehicle handling without requiring trajectory planning. Note that two additional steering inputs were included in the system, which assisted the driver to better handle the driving experience via rapid and high maneuverability in the context of the four-wheel-independent-steering (4WIS) mode.

For a moving vehicles, maintaining pure rolling is essential because wheel slip may impart lateral forces on the wheels, decreasing the stability and consuming additional driving energy [37], [38]. Most mechanical steering arrangements turn the front wheels using a hand-operated steering wheel that is positioned in front of the driver via the steering column, universal joints, rack and pinion and a steering tie rod. The existence of mechanical linkages among wheels guarantees that the kinematic constraint minimizes wheel slip in traditional EVs, as shown in Fig. 1. Nevertheless, the removal of the mechanical linkage among wheels allows each wheel move independently, which means that steering control is more difficult in a 4WIS EV.

Despite the success of the existing approaches/systems for the steering system of a 4WS4WD EV, a variety of factors still challenge researchers. Much research has been conducted on the steering system of the space-saving 4WS4WD EV, which has focused on the following two primary problems: 1) How can two front wheel steering (2FWS) and OIS be integrated to easily handle EVs during both high-speed driving and high-maneuverability turning, such as ZRT and LP motion? 2) How to realize the control strategy for a space-saving steering system of a 4WS4WD EV with higher maneuverability and flexibility?

To tackle these problems, the most critical issue is how to realize the design of the OIS for the 4WS4WD EV. In this paper, we introduce a design and control strategy for an OIS EV in an integrated solution with a retrofitted steering system. The 2FWS

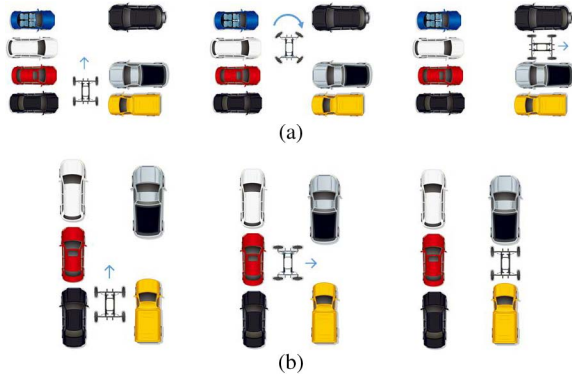


Fig. 2. Need for parking in a limited area or congested streets. (a) Progression of zero radius turning in reverse parking. (b) Progression of lateral motion in lateral parking.

mechanism of our 4WS4WD EV is designed to control the front wheels via a redesigned steer system for vehicle traveling at high speeds. Additionally, a novel OIS mechanism is proposed to achieve an angle range of $-35^\circ \sim +90^\circ$. When driving in a limited parking place or congested streets in crowded cities, particularly for the parking scenarios shown in Fig. 2, the driver can control the OIS to realize ZRT and LP motion to simplify the parking process in narrow spaces. Finally, the control strategy for the space-saving steering system of an EV is redefined for the integrated 2FWS and OIS, which can easily handle EVs for high-speed driving or high-maneuverability turning.

The rest of the paper is organized as follows. In Section II, the general system architecture of the novel steering system for the space-saving 4WS4WD EV is described, and its main components are described. Section III focuses on the integrated design of the OIS system for the space-saving 4WS4WD EV. The control strategy and related electric design are developed in Section IV. The modeling and analysis of the proposed system are presented in Section V. The system simulation and verification are reported in Section VI. Finally, several conclusions are provided in Section VII.

II. SYSTEM ARCHITECTURE

The general architecture of our system is shown in Fig. 3, which presents the OIS mechanism integrated steer-by-wire and the control strategy for the space-saving steering system of an EV.

In the first part, when the vehicle is traveling at a high speed, the redesigned 2FWS mechanism controls the front wheels to realize conventional steering motion. The two front wheels can be steered in pairs and independently due to the redesigned steering tie rods. The driver can control the front wheels in the traditional way by means of a steering wheel via the steering column, universal joints, rack and pinion and the redesigned steering tie rods.

The second part is the OIS mechanism integrated steer-by-wire. When a vehicle is travelling in a limited parking place or a congested street, where high maneuverability is required, the driver can control the 4WS4WD EV through the proposed OIS mechanism. The realization of omni-directional motions

depends on the design of the OIS mechanism. The wheel is driven independently by a brushless permanent magnet DC in-wheel motor, which is controlled by the central controller. Thus, all the wheels can be steered independently and driven independently by wire. The driver can control the OIS to turn the four wheels independently with the assistance of the steer-by-wire technologies. Our OIS mechanism can achieve an angle range of $-35^\circ \sim +90^\circ$, which is a solid basis for ZRT and LP motion.

The final part is the control strategy for the space-saving steering system of an EV, which can easily handle the EV for high-speed driving or high-maneuverability turning, such as ZRT and LP motion. The central controller obtains an omni-directional motion control command from the steering buttons, and the central controller will turn the four wheels independently via the steering motor, reconstructed rack and pinion and steering arm. As shown in the middle image of Fig. 3, the control strategy can choose the steering model for conventional steering at high speeds or the OIS model for ZRT or LP. After the control strategy is implemented, the maneuverability and flexibility of the proposed space-saving steering system 4WS4WD EV can be achieved.

III. DESIGN OF A NOVEL OMNI-DIRECTIONAL STEERING FOR 4WS4WD EV

The purpose of the proposed OIS system is to attain better maneuverability and flexibility. As introduced above, the conventional mechanical steering system is an extensively researched technology, deeply embedded in the control system. However, the OIS system improves the steering angle and range. Considering the advantage of the conventional mechanical steering in stability and the promotion of 4WS and x-by-wire in flexibility, we make full uses of conventional mechanical steering technologies and OIS technologies to build an integrated OIS system for a space-saving 4WS4WD EV.

A. CAD Modeling

The advantage of conventional mechanical steering technology is that the kinematic constraint is maintained by the mechanical linkages between the two front wheels. The stabilities of traditional vehicles are good, which is a safeguard for omni-directional vehicles. In a four-wheel independent steering vehicle, each wheel is steered independently, which makes steering control extremely difficult. Although each wheel can independently achieve the kinematic constraint in steady state using a typical position controller, kinematic constraint cannot be guaranteed in transient states.

We propose a redesigned 2FWS mechanism to cope with the wheel misalignment problem, as shown in Fig. 4. The mechanism consists of a steering wheel, steering column, universal joints, rack and pinion and steering tie rods. The steering column is powered by a human driver through the steering wheel. Universal joints are used to connect the steering column with the pinion shaft, rack and pinion, and redesigned steering tie rods. The redesigned steering tie rods allow the two front wheels to be coupled for better stability and independent for better maneuverability through a solenoid valve, which

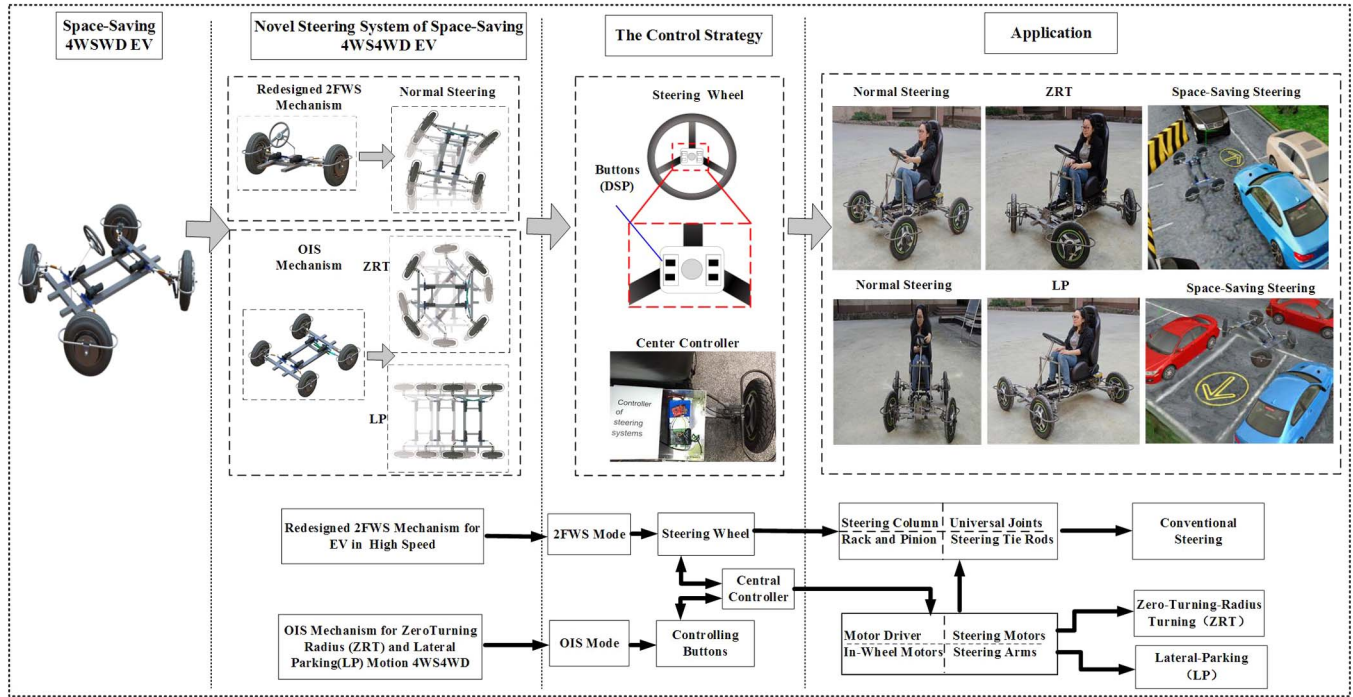


Fig. 3. Flowchart of the proposed omnidirectional steering system.

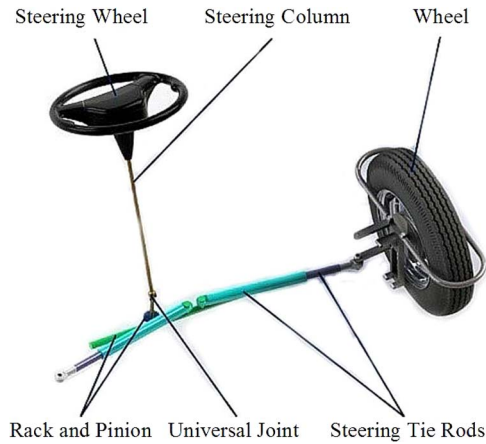


Fig. 4. Structure of the redesigned 2FWS mechanism.

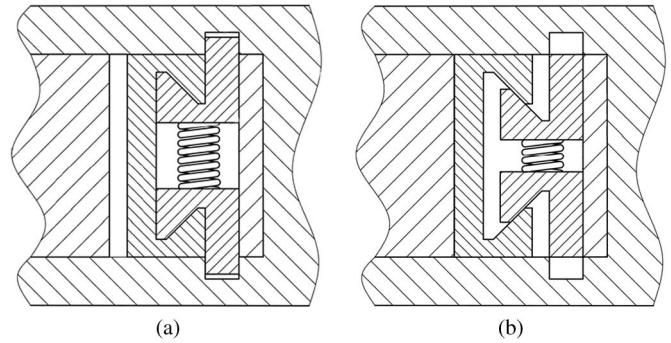


Fig. 5. Design and schematic diagram of the solenoid valve. (a) Power off. (b) Power on.

is shown in Fig. 5. The two parts of the steering tie rods work as one when the solenoid valve is locked, whereas they are separated when the solenoid valve is released. When the solenoid valve is locked, the steering motor serves as powered steering assistance. The solenoid valves are designed to be locked when the power is off.

As shown in Fig. 6, in 2FWS mode, drivers maneuver steering wheel and the two front OIS mechanisms work as power-assisted steering. The solenoid valves are locked while the steering column remains rigid. When a driver is steering in 2FWS mode, the system obtains the signals of the steering angle and then controls the steering motors that turn the wheels.

The proposed OIS system consists of four sets of novel OIS mechanisms that depend on steer-by-wire and drive-by-wire technologies. Fig. 7 illustrates the overview of the proposed OIS system, where each set of the OIS mechanisms is installed

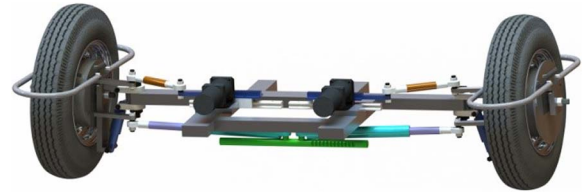


Fig. 6. Overview of the steering system integrated 2FWS mode and OIS mode.

on the chassis and support the vehicle body by contacting the ground.

The realization of omni-directional motions depends on the OIS mechanisms, as shown in Fig. 8. Each set of OIS mechanisms is inclusive of a DC geared motor for steering, a driving pinion that is driven by the output shaft of the DC geared motor, a redesigned rack that is used to transfer the rotational motion into linear motion and a steering arm, which was developed to couple the redesigned rack with the wheel. The wheel is driven independently by a brushless permanent magnet DC in-wheel

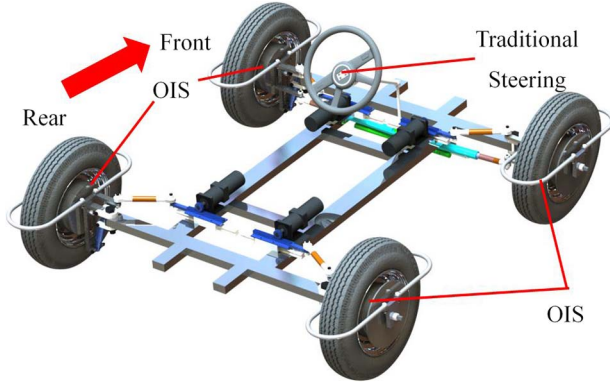


Fig. 7. Four sets of OIS mechanisms are connected to the chassis, and the traditional mechanical steering system is located laterally to the OIS mechanisms in the front of the chassis.

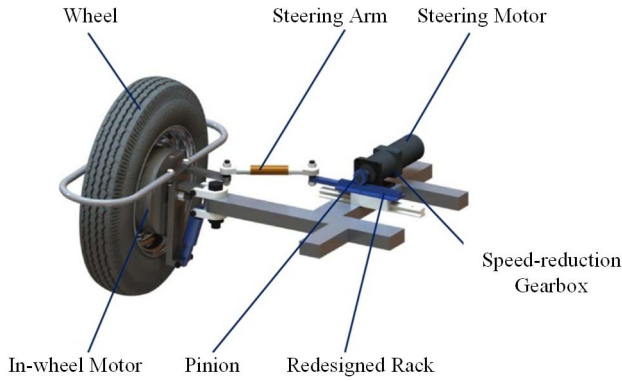


Fig. 8. Specific CAD configuration of an OIS.

motor, which is controlled by the central controller. Thus, all the wheels can be steered independently and driven independently by wire. Omni-directional motions, such as ZRT and LP, can be achieved via cooperation of the four steering motors and four in-wheel motors.

The OIS mechanism is designed to achieve an angle range of $-35^\circ \sim +90^\circ$, which is a solid basis for ZRT and LP motion. As shown in Fig. 8, the redesigned rack and wheels are linked by the steering arm link with universal joints. This is a new approach to absorb the vertical dumping motion of the wheels, which reduces roadway distractions for additional control. Furthermore, the steering arm, which is attached to the rack and wheel, maximizes the turning angle range and simplifies the control to one linear motion of the rack. Mechanical linkages are coupled between two steering wheels that can achieve kinematic constraint while vehicles are driving. In comparison, steer-by-wire vehicles retain the constraint by precise steering motor control. Note that when vehicles are traversing complex terrain, precise control is difficult to attain. If vehicles experience strong shocks that are random and unpredictable, it is even harder for the steer-by-wire system to function.

B. Proposed Prototype Vehicle

Based on the CAD design, we manufactured all of the parts, including the redesigned 2FWS mechanism, OIS mechanisms



Fig. 9. Overview of the prototype EV.

TABLE I
DETAILED MEASUREMENTS OF THE PROPOSED PROTOTYPE

Size	$L1 \times L2 \times H1$	$1.65 \text{ m} \times 1.50 \text{ m} \times 1.45 \text{ m}$	External dimension
Chassis height	H2	0.2 m	Gap between the chassis and ground
Wheel base	L3	1.10 m	Distance between the wheel shafts
Wheel diameter	D	0.45 m	External diameter of the wheels
Total weight	M		Exclusive of driver
Driving power	P1	800 W*4	Independent for the four wheels
Steering motor	P2	60 W*4	OIS and power steering
Gear ratio	i	64	Increased torque
Battery	W	12 Ah*4	Four batteries in series

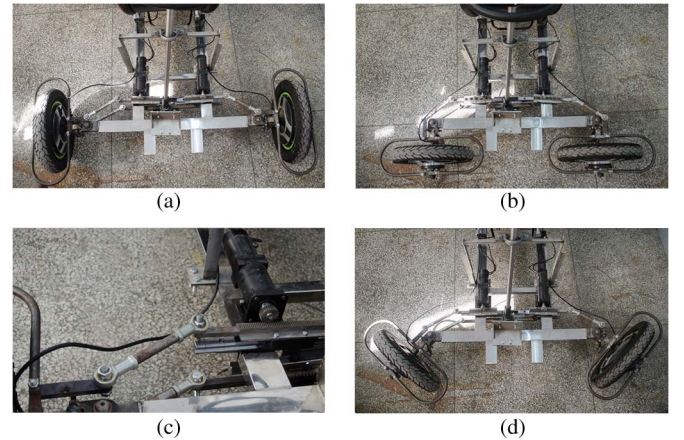


Fig. 10. (a) Overview of the steering system in the front wheels, where the OIS mechanism is on top and the 2FWS is at the bottom. (b) Positions of steering systems when the vehicle is in LP mode. (c) Posture of the steering systems when the vehicle is in ZRT mode. (d) Features of the steering system.

and seat configuration. Then, we assembled a full-size prototype vehicle, as shown in Fig. 9. The prototype is assembled with one seat for the driver and contains fully functional parts, as marked in the figure. The measurements are also presented in the figure. The detailed measurements are shown in Table I. Fig. 10 shows an overview of the space-saving 4WS4WD EV steering system of our prototype.

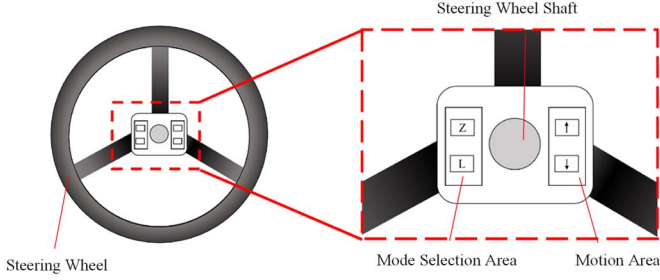


Fig. 11. Steering wheel and control panel. The control panel contains two areas beside the central shaft of the steering wheel.

IV. CONTROL STRATEGY FOR OMNI-DIRECTIONAL STEERING

A. Intuitive Interface for Control

To satisfy the research need of handling the OIS for the space-saving steering system 4WS4WD EV, the conventional steering wheel is no longer appropriate. Discarding the common mode-changing interface of the OIS, we propose an intuitive interface for customized control of this omni-directional steering. As shown in Fig. 11, the interface system consists of a steering wheel for conventional steering at high speeds and buttons for the OIS, such as ZRT and LP. By intuitive pressing, the driver can easily handle the omni-directional motion, leaving any high-load calculations and attitude planning to the central controller. The panel is installed in the center of the steering wheel and divided into two areas, namely, the mode selection and motion areas. In the mode selection area, button “Z” represents ZRT mode and “L” represents LP mode. The control system resets the position of the vehicle when “Z” or “L” is touched twice in succession. In the motion area, there are the UP and DOWN buttons. When pressing UP or DOWN button, the vehicle moves in the direction of the arrow. The intrinsic logic for the UP and DOWN buttons is similar to the turning light lever. Once the button is released, the vehicle will stop. This driving operation is specifically designed for the space-saving 4WS4WD EV because omni-directional motion requires low speeds and a fixed mode.

When driving and parking in a crowded environment, the driver pushes one button for ZRT or LP, and then, the solenoid valve unlocks; the steering wheel acts as an input to the acceleration controller. The space-saving 4WS4WD EV turns in the same direction as the driver drives the steering wheel for ZRT or moves to the same side in LP as the steering wheel turns. The scheme is illustrated in Fig. 12 with examples. In LP or ZRT, the driver selects a mode in the mode selection area as the first step. Then, the driver holds the UP or DOWN button to conduct a leftward motion or rightward motion in LP mode or clockwise turning or counterclockwise turning in ZRT mode. If the motion button is released, the vehicle stops. The driver can easily adjust the vehicle position by pressing the motion button repeatedly. The goal of this intuitive interface is to avoid confusing the driver in omni-directional motion and to reduce the learning cost without extra training. Furthermore, this control strategy is efficient and effective when implemented

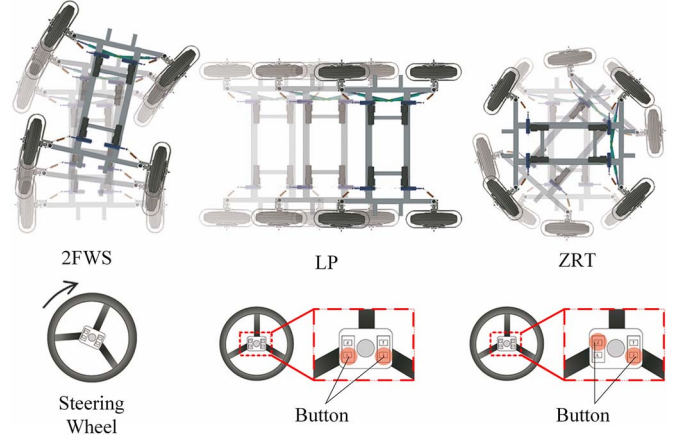


Fig. 12. Three steering modes of OIS vehicles. (a) In 2FWS mode, drivers handle the steering wheels as usual. (b) In LP mode, drivers press the “L” button in the mode selection area and hold the DOWN button in motion area when rightward LP is needed. (c) In ZRT mode, drivers press the “Z” button in the mode selection area and hold the DOWN button in the motion area when clockwise turning is needed.

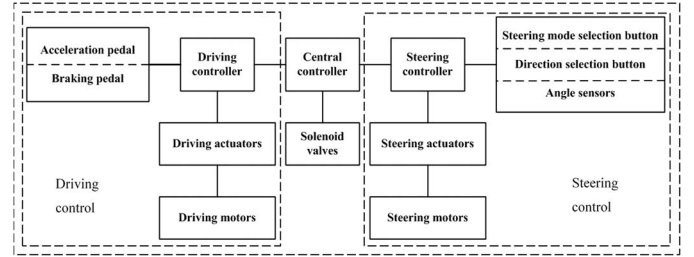


Fig. 13. Outline of the electric control system.

in software and hardware, which avoids any additional input or trajectory planning benefitted by the integrated steering system.

B. Control Realization

The proposed OIS system evolved from the traditional steering system. It is a steer-by-wire and drive-by-wire system. To realize these functions in both omni-directional and conventional modes, an electric control system is necessary for the space-saving 4WS4WD EV system. Fig. 13 shows the outline of the proposed electric control system. There are three parts in the system, i.e., a central controller, which is the kernel of the control system, a driving controller and a steering controller.

When the driver starts the vehicle, the drive controller obtains voltage signals from the acceleration pedal and converts the voltage signal to a speed control signal. Next, the speed control signal is sent to the central controller, and the steering controller obtains voltage signals from the steering interface and angle sensors and converts the voltage signals to steering control signals. Then, the steering control signals are also sent to the central controller. The central controller will make an operational judgment based on the speed control signal and steering control signals. Then, the drive controller and the steering controller will control the four driving motors and four steering motors through actuators to achieve different operations depending on the particular judgment. The different operations primarily

TABLE II
OPERATING STATE OF DRIVING MOTORS

		Conventional motions		Omni-directional motions	
		Straight	Turning	ZRT	LP
Solenoid valves		Lock	Lock	Unlock	Unlock
Driving motors	Front Left	S-F	D-F	S-F	S-F
	Front Right	S-F	D-F	S-B	S-B
	Rear Left	S-F	D-F	S-F	S-B
	Rear Right	S-F	D-F	S-B	S-F
Steering motors	Front Left	N	C	C	C
	Front Right	N	C	U	U
	Rear Left	N	N	C	C
	Rear Right	N	N	U	U

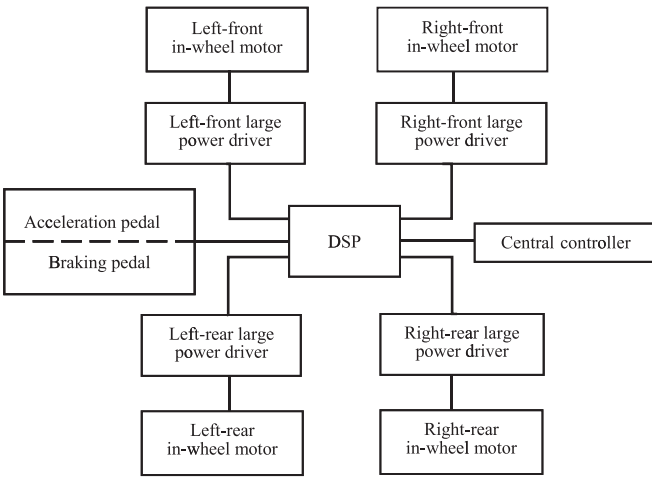


Fig. 14. Hardware view of the proposed OIS steering system.

consist of conventional motions and omni-directional motions. The operating states of the four driving motors, four steering motors and solenoid valves under the different motion states are shown in Table II. “S,” “D,” “F” and “B” denote the same speed and, the differential, forwards and backwards motions of the driving motors, respectively. “N,” “C” and “U” denote stop, clockwise rotation and counterclockwise rotation of the steering motors, respectively.

Fig. 14 shows the hardware view of the control, which is based on the digital signal processor (DSP) controller. The processor of the driving system has six pulse width modulation (PWM) output ports to command the proposed four in-wheel motors and communicate with the central controller. The selected DSP provides the I/O ports for signal transmission with the A/D modular to convert the analog signals of the acceleration into digital signals. The DSP driving controller primarily works to obtain voltage analog signals of the acceleration, which then converts the signals into speed control digital signals that are sent to the central controller. On the other side, the DSP receives control signals of speed, direction and torque and sends the corresponding command to the motor drivers. In this way, the DSP achieves driving control of the omni-directional vehicles. An overview of the central controller is shown in Fig. 15.



Fig. 15. Hardware view of the proposed OIS steering system.

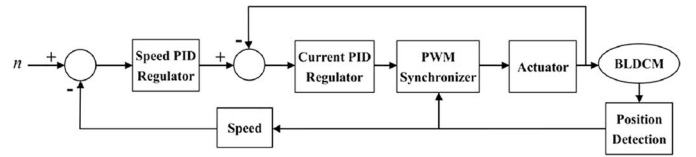


Fig. 16. Flowchart of the speed and current double closed-loop PID control system.

TABLE III
PARAMETERS OF THE ZIEGLER–NICHOLS METHOD

Type of controller	K_p	T_i	T_d
P	$0.5K_u$	∞	0
PI	$0.45K_u$	$T_u/1.2$	0
PID	$0.6K_u$	$0.5 T_u$	$0.125 T_u$

The speed control and response are keys technologies that impact the stability of the vehicle. To achieve high-precision speed control and response of the brushless DC in-wheel motor, a speed and current double closed-loop proportion integration derivative (PID) control system was designed in the simulation. The speed closed-loop and the current closed-loop are controlled by a PID regulator. We established a driving control system simulation model using Matlab/Simulink. The system depends on the speed and current double closed-loop PID control system, as shown in Fig. 16. The simulation parameters of the brushless DC in-wheel motor are as follows: the rated power is 1 kW, and the rated voltage is 48 V. Ziegler-Nichols method was chosen to regulate the parameters of PID, the detailed setting is shown in Table III.

Where in Table III, K_p represents proportion coefficient, T_i represents integration coefficient, and T_d represents integration coefficient. K_u represents the ultimate gain and T_u represents the oscillation period.

V. MODELING AND ANALYSIS

A. Kinematics Model of the OIS Steering Structure

The design of the OIS is developed based on the moving theory of four bar linkages. In the following, we present a

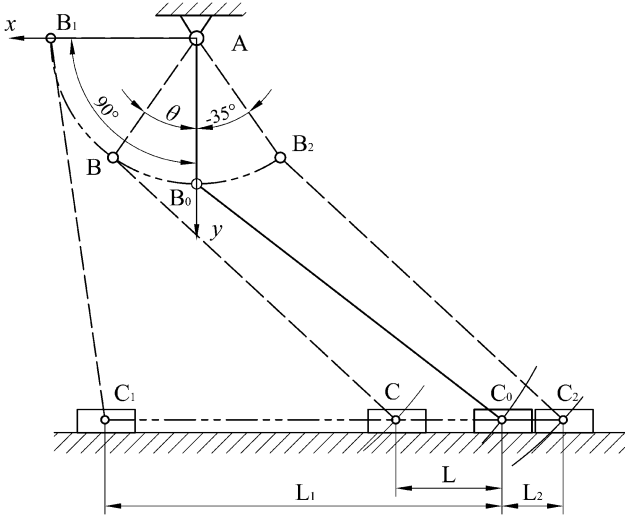


Fig. 17. Kinematics modeling of the omnidirectional independent steering structure.

graphical design of the omni-directional steering structure. A simplified schematic is shown in Fig. 17.

The rotating angle of lever AB represents the steering angle of the wheel. The role of lever BC is the same as the steering arm. The linear motion of the slider is regarded as the redesigned rack. The rotating angle of lever AB is confined to between -35° and $+90^\circ$ to meet the demands of the omni-directional motions and conventional motions. We preset the length of the lever BC to be equal to the steering arm.

To obtain the linear motion information of point C, a coordinate system is established as shown in Fig. 17, and the origin of the coordinate is located at point A. Then the coordinates of point B can be derived by:

$$x_B = l_{AB} \sin \theta, \quad y_B = l_{AB} \cos \theta, \quad -35^\circ \leq \theta \leq +90^\circ \quad (1)$$

where l_{AB} is the length of lever AB, and θ is the rotating angle of lever AB.

Because the vertical distance between point A and point C is constant y_C , the length of lever BC is constant. Then, the abscissa x_C of point C is attained by:

$$(x_C - x_B)^2 - (y_C - y_B)^2 = l_{BC}^2 \quad (2)$$

$$(x_C - l_{AB} \sin \theta)^2 + (y_C - l_{AB} \cos \theta)^2 = l_{BC}^2$$

$$\Rightarrow x_C = l_{AB} \sin \theta - \sqrt{l_{BC}^2 - (y_C - l_{AB} \cos \theta)^2} \quad (3)$$

Thus, we can obtain the displacement of point C, i.e., the linear motion of the slider:

$$L = x_C - x_{C0} \quad (4)$$

where x_{C0} is the abscissa of point C when the rotating angle θ of lever AB is zero.

Then, we can obtain the relationship between the rotating angle of lever AB and the linear motion of the slider, i.e., the steering angle of the wheel and the linear motion of the rack. Because the rack is driven by the driving pinion, the

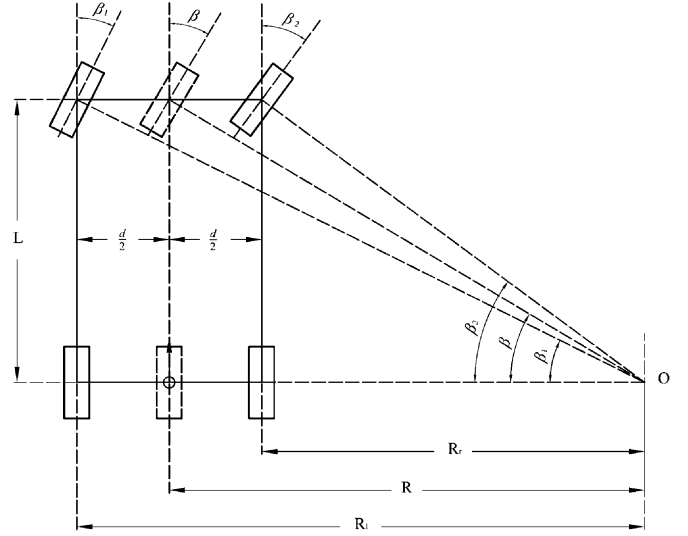


Fig. 18. Steering angle and velocity of each wheel under front-wheel steering mode.

relationship between the steering angle of the wheel and the angular displacement of the driving pinion can be acquired based on the rack and pinion transmission. The transmission ratio i is:

$$i = \frac{v}{\omega} = \frac{vt}{\omega t} = \frac{l}{\theta_p} = r \quad (5)$$

where v and l represent the linear speed and distance of the rack, respectively, and ω , θ_p and r represent the rotational speed, angular displacement and the radius of the reference circle of the driving pinion, respectively.

We can obtain the rotational angle θ_m of the steering motor depending on the reduction ratio i_r of the speed-reduction gearbox:

$$\theta_m = i_r \theta_p = \frac{i_r}{r} \left(l_{AB} \sin \theta - \sqrt{l_{BC}^2 - (y_C - l_{AB} \cos \theta)^2} + \sqrt{l_{BC}^2 - (y_C - l_{AB} \cos \theta)^2} \right) \quad (6)$$

Thus, we can achieve steering wheel control through the steering motor control depending on (3)–(6).

B. Kinematics Analysis of the Different Steering Modes

1) *Analysis of the 2FWS System Mode:* All of the axes of the wheels must intersect at the instantaneous center of rotation to avoid side-slipping when the vehicle turns under the front wheel steering mode by Ackerman's law. The relationship of the steering angle and velocity of each wheel is shown in Fig. 18.

When the steering angle is assumed to as β , the instantaneous center of rotation can be located by:

$$R = \frac{L}{\tan \beta} \quad (7)$$

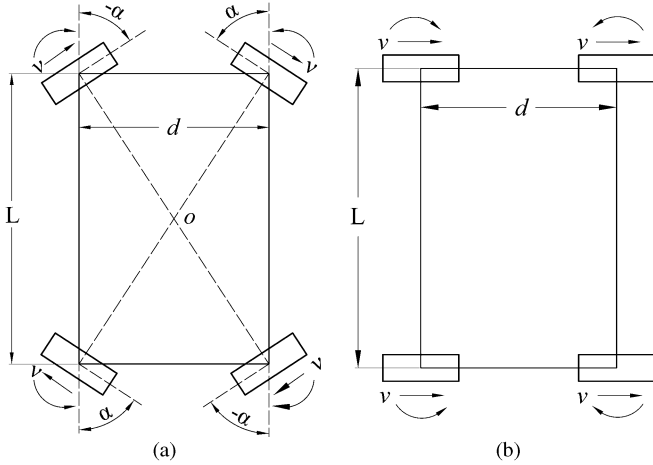


Fig. 19. Steering angle of each wheel in OIS mode. (a) ZRT mode. (b) LP mode.

where L is the axis base. The left and right rear wheel turning radius R_l and R_r can be derived by

$$R_l = R + \frac{d}{2} = \frac{L}{\tan \beta} + \frac{d}{2} \quad (8)$$

$$R_r = R - \frac{d}{2} = \frac{L}{\tan \beta} - \frac{d}{2} \quad (9)$$

where d is the wheel base. The front left steering angle β_1 is attained by

$$\beta_1 = \arctan\left(\frac{L}{R_l}\right) = \arctan\left(\frac{L}{\left(\frac{L}{\tan \beta} + \frac{d}{2}\right)}\right) \quad (10)$$

The front right steering angle β_2 is attained by

$$\beta_2 = \arctan\left(\frac{L}{R_r}\right) = \arctan\left(\frac{L}{\left(\frac{L}{\tan \beta} - \frac{d}{2}\right)}\right). \quad (11)$$

When the velocity of the vehicle is assumed to be v , we can deduce the velocity of each wheel:

$$v_{fl} = v \frac{R_l}{R \cos \beta_1}, v_{fr} = v \frac{R_r}{R \cos \beta_2}, v_{rl} = v \frac{R_l}{R}, v_{rr} = v \frac{R_r}{R} \quad (12)$$

where v_{fl} represents the velocity of the front left wheel, v_{fr} represents the velocity of the front right wheel, v_{rl} represents the velocity of the rear left wheel, and v_{rr} represents the velocity of the rear right wheel.

Accordingly, when the driver turns the front wheels to the target angle requested by the steering wheel, the two front steering motors and four driving motors are controlled based on (7)–(12).

2) *Analysis of the OIS Mode:* To attain a higher maneuverability of the vehicle, the proposed OIS system requires a novel control strategy for the omni-directional steering mode. The two solenoid valves are unlocked under the omni-directional steering modes, and the two front wheels can be steered independently. There are two modes of OIS motion, namely, ZRT and LP. As shown in Fig. 19(a), each wheel of the vehicle is

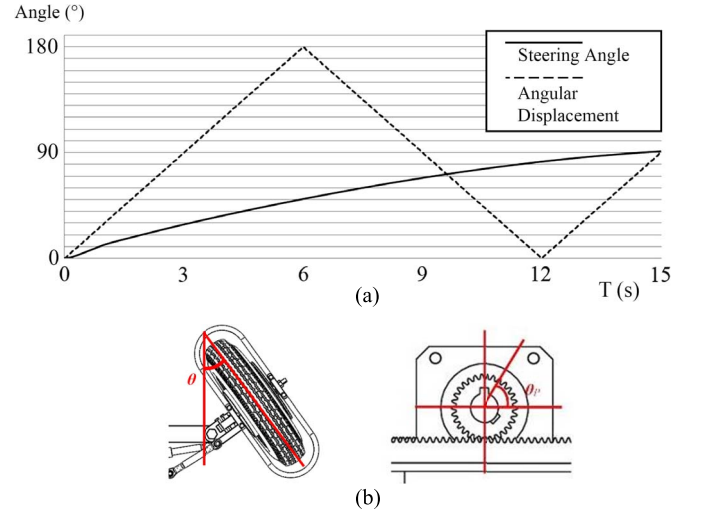


Fig. 20. Simulation of the mechanical motion. (a) Relationship between steering angle θ and angular displacement θ_p . (b) Steering angle implies the turning angle of wheels. The angular displacement is the included angle of the present position and the start position of gears on the steering motor, which remain less than 180° .

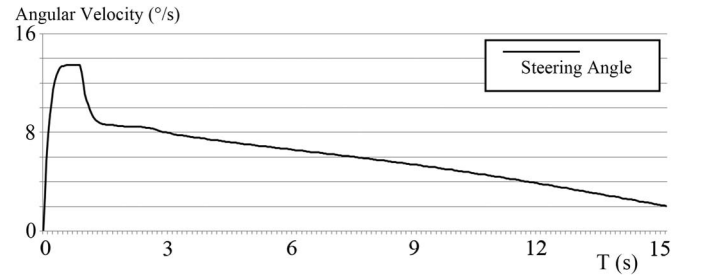


Fig. 21. Angular velocity of the steering wheel.

TABLE IV
TURNING PARAMETERS OF THE VEHICLE

Steering mode	Steering angle ($^\circ$)		Speed of in-wheel motor (r/min)			
	β_1	β_2	n_{mfl}	n_{mfr}	n_{mrl}	n_{mrr}
2FWS	9.138	11.039	166	138	164	135
4WS	8.412	12.315	180	124	180	124

steered independently until its wheel axle points to the geometric center of the vehicle. The right front wheel and left rear wheel are steered to the same positive angle $\alpha = \arctan\left(\frac{L}{d}\right)$, and the other two are steered to the same negative angle $-\alpha$. The left front and rear wheels are driven forward, and the other two are driven backwards. The speeds of all the wheels are numerically the same. The vehicle can rotate around the geometric center of the vehicle without translation, and the turning radius is zero. Hence, we call it ZRT motion. When α increases to 90° , the four velocity vectors are parallel and have the same length without a side effect, as shown in Fig. 19(b). The vehicle can accomplish LP motion.

VI. SIMULATION AND EXPERIMENTAL VERIFICATION

The goal of this section is to demonstrate experimentally and scientifically the validity of our proposed steering system

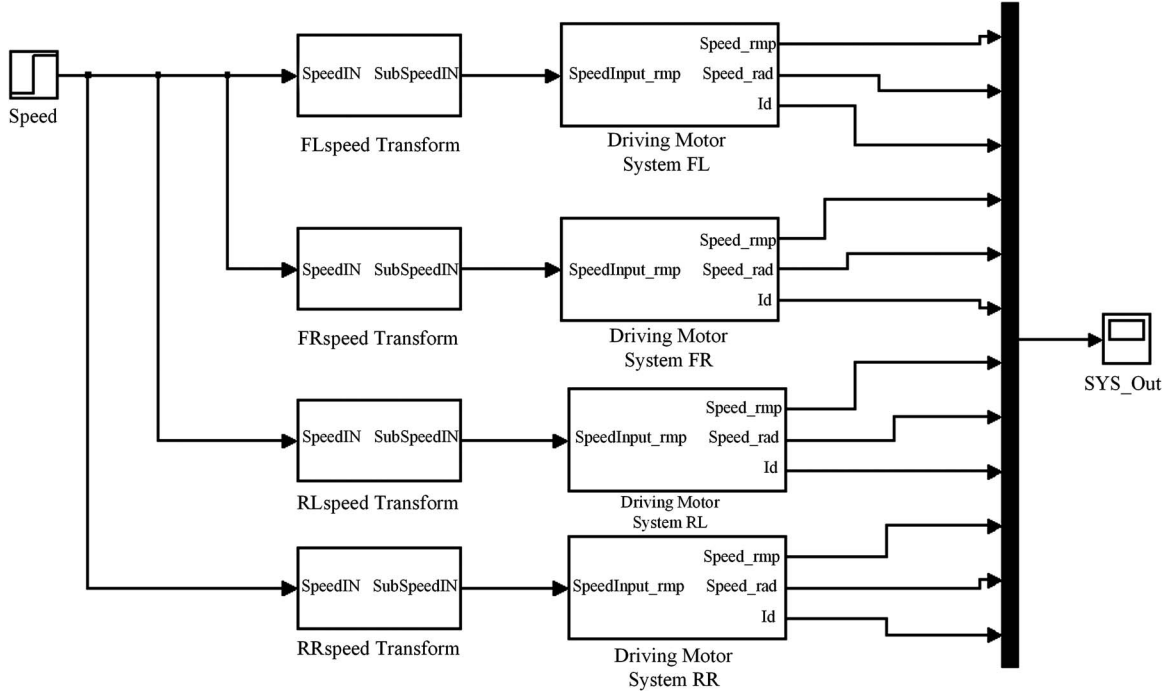


Fig. 22. Driving control system simulation model.

of a space-saving 4WS4WD EV that is characterized by its higher maneuverability and flexibility. It is extremely important to evaluate the performance of the mechanical transmission and motor drive. To validate the above analysis and provide a valuable reference for the following project design, our proposed steering system of the space-saving 4WS4WD EV was tested adequately in both simulation and experimental environment.

A. Simulation of the Mechanical Motion

The simulation of the mechanical motion was implemented using *Solidworks* motion analysis. In the simulation, we chose a set of omni-directional independent steering mechanisms as the subject of research to study the relationship between the steering angle θ of the wheel and the rotational angle θ_m of the steering motor. We used a constant-speed revolved motor to simulate the rotation of the driving pinion. The relationship between the steering angle θ of the wheel and the angular displacement θ_p of the driving pinion is shown in Fig. 20, and thus, we can obtain the corresponding rotational angle θ_m of the steering motor when we determine the steering angle θ based on (6). This provides a valuable reference for our future works. Additionally, we can obtain the angular velocity of the steering wheel, as shown in Fig. 21. The angular velocity changes rapidly in the first 1.2 seconds and then becomes smooth. Consequently, the OIS mechanism receives an impulse in the beginning, and the stability of the steering operation is affected. Therefore, the low angular velocity of the steering motor is necessary in the beginning.

To ensure the feasibility of omni-directional motions with the design structure, a verification experiment was performed based on a 3D model platform. The functions of omni-directional motions, i.e., ZTR motion and LP motion, can be realized with four sets of OIS mechanisms, which would achieve good

performance in limited parking spaces or congested streets in terms of maneuverability and flexibility.

B. Simulation of the Driving Motors

In our proposed OIS system configuration, each wheel can be driven by a brushless DC in-wheel motor independently. As a result, the velocity of each wheel can be controlled independently. The independent control of the velocity is beneficial for optimizing the driving efficiency and improving vehicle stability. The vehicle stability is closely related to the ability to achieve a given curvature. To follow a given curvature, the velocity of each wheel must be equal to certain values according to (12). The velocity of the wheel is determined by the speed of the in-wheel motor:

$$v_w = 2\pi R n_m \Rightarrow n_m = \frac{v_w}{2\pi R} \quad (13)$$

where v_w represents the velocity of the wheel, R is the radius of the wheel, and n_m represents the speed of the in-wheel motor.

We assume that the velocity of the vehicle is 20 km/h before turning, the velocity reduces to 15 km/h when turning, and the steering angle β is assumed to be 10 degrees. The radius of wheel is 265 mm, and thus, we can obtain the reference speed of the in-wheel motor. The speed of the in-wheel motor is calculated as 200 r/min before turning and is 150 r/min when turning. The axis base of the vehicle with our proposed OIS system is 1100 mm, and the wheel base is 1200 mm. Thus, the turning parameters are determined based on the conventional steering mode, as shown in Table IV. The turning speed is set to 5 km/h under the OIS mode, whereas the speed of the four in-wheel motors is 50 r/min. n_{mfl} represents the speed of the front-left driving motor, and n_{mfr} represents the speed of the front-right driving motor. n_{mrl} represents the speed of the

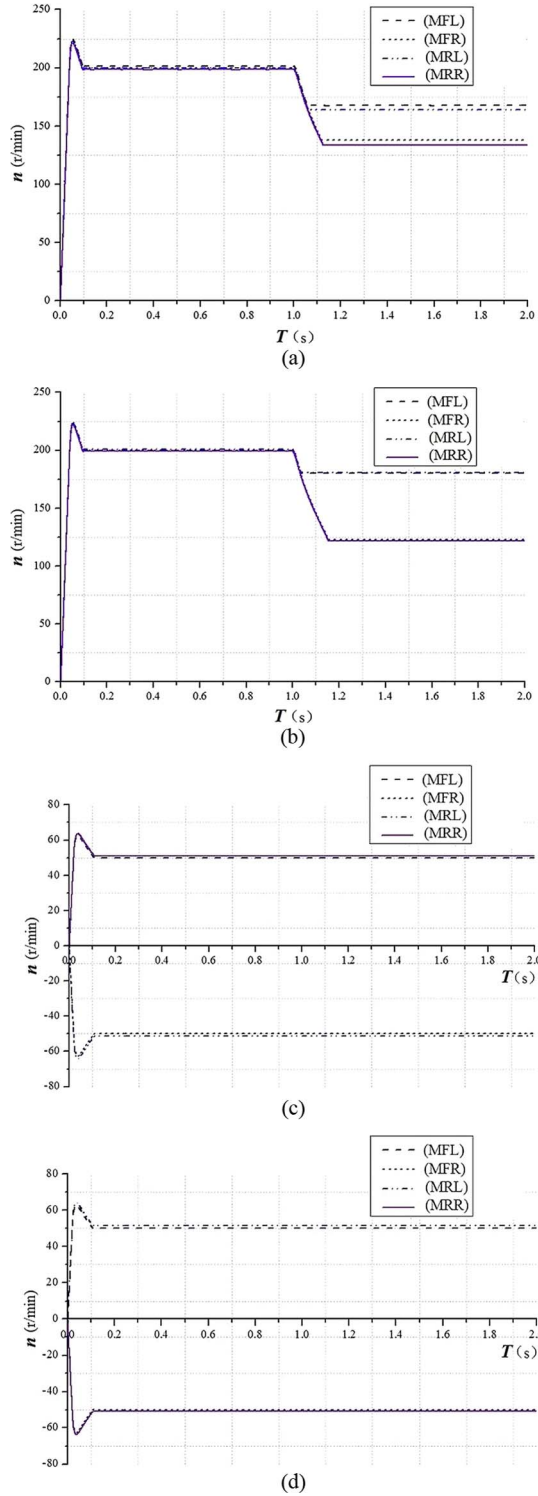


Fig. 23. Simulation results of the driving control system. MFL, MFR, MRL, and MRR represent the motor in front-left, front-right, rear-left, and rear-right, respectively.

rear-left driving motor, and n_{mrr} represents the speed of rear-right driving motor.

We ran the simulation model in *Matlab* software, as shown in Fig. 22. Then, we obtained the simulation results of the different steering modes. Fig. 23(a) shows the simulation results of differential control in the two-front-wheel steering mode. Fig. 23(b) shows the simulation results of differential control in

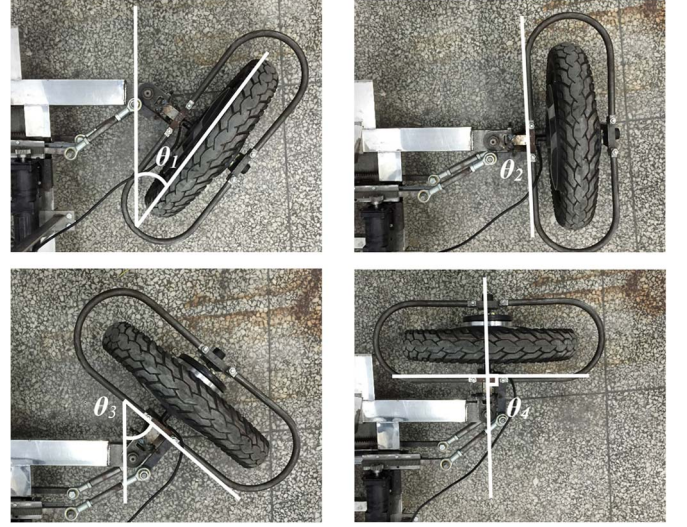


Fig. 24. Steering experiment of the proposed prototype of the OIS mechanisms.

the four-wheel steering mode. Fig. 23(c) shows the simulation results of speed control in the ZRT mode. Fig. 23(d) shows the simulation results of speed control in the LP mode. The simulations in Fig. 23 demonstrate the good performance of the proposed driving control system. According to Fig. 23(a) and (b), the system response time of the driving motors is 0.1 seconds when the speed is increased from 0 to 200 r/min, and the peak speed is 220 r/min. The deceleration control system response of all the driving motors is rapid and smooth. As shown in Fig. 23(c) and (d), the system response time of the driving motors is 0.1 seconds when the speed is increased from 0 to 50 r/min, and the peak speed is 65 r/min. The acceleration control system response time of all the driving motors is 0.1 seconds. The driving control system has good dynamic and steady-state characteristics.

C. Steering Experiment of the Novel OIS Mechanisms

To verify the feasibility of the designed systems, we processed and assembled a set of OIS mechanisms after completing the design and simulation. Then, the steering experiment was performed based on the prototype, as shown in Fig. 24. The experimental results show that the proposed novel OIS mechanism can realize $-39^\circ \sim +92^\circ$ steering angles and thus, meets the design goal of the steering system. We also obtained the relationship between the specially appointed wheel steering angle and the displacement of the redesigned rack. The results are shown in Table V.

We also measured the angles when in progress, as shown in Fig. 24. The largest measured turning angle θ_1 for 2FWS was -39° . θ_2 remains zero when driving in a straight line. θ_3 is the steering angle when the vehicle is in ZRT motion, which was measured as 49° . When the vehicle was in LP mode, the steering angle θ_4 was 90° . Figs. 25 and 26 demonstrate the potential applications of the proposed space-saving 4WS4WD EV.

The conventional steering system turned in a range of $-35^\circ \sim +35^\circ$, where the steering wheel rotates between $-180^\circ \sim +180^\circ$. The prototype achieved ZRT turning when the four wheels turned by 48° . Additionally, the OIS system

TABLE V
RELATIONSHIP BETWEEN WHEEL STEERING ANGLE AND THE
DISPLACEMENT OF THE REDESIGNED RACK

Steering angle (°)	-39	-35	-25	-15	-5	0
Displacement (mm)	-12.4	-11.7	-10	-7	-2.5	0
Steering angle (°)	+5	+15	+25	+35	+45	+90
Displacement(mm)	2.6	8.7	15.5	22	30	62

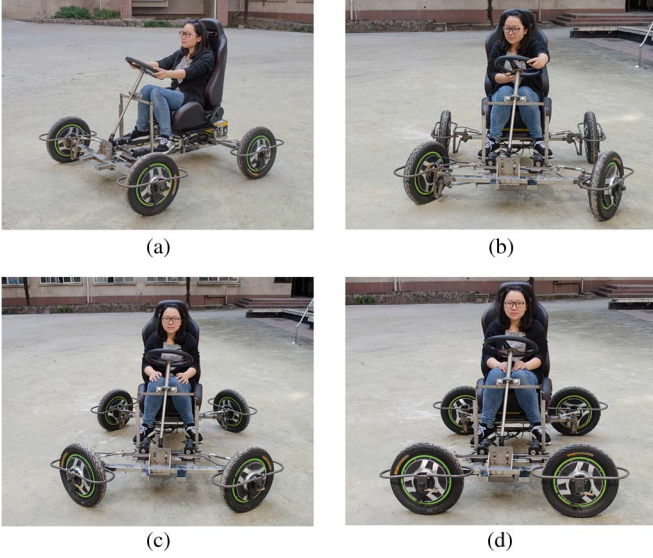


Fig. 25. Prototype experiment of the proposed OIS 4WS4WD EV. (a) Straight line. (b) Normal turning. (c) ZRT motion. (d) LP motion.

accomplished LP motion when all four wheels were turned 90 degrees.

To validate the proposed steering system, we conducted four modes of steering on our prototype on campus roads. The prototype was maneuvered at engaged angles, and all of the designed functions of the steering system were realized, including normal turning, ZRT turning, and LP turning, as shown in Figs. 25 and 26.

VII. CONCLUSION

In this paper, we present the design, modeling, simulation and road tests of a novel steering system for a space-saving 4WS4WD EV with high maneuverability and flexibility. The proposed system consists of an improved 2FWS mechanism, an OIS mechanism integrated steer-by-wire and the control strategy of the space-saving steering system. The 2FWS mechanism of the proposed 4WS4WD EV is designed to control the front wheels via the redesigned steering system when the vehicle is traveling at a high speed. The driver can control the vehicle in the traditional way through the redesigned 2FWS mechanisms, where two sets of OIS mechanisms provide power-assistance steering. The redesigned 2FWS mechanisms can guarantee the stability of the vehicle due to mechanical linkages between the wheels. The proposed OIS mechanism achieves an angle range of $-35^\circ \sim +90^\circ$, which is a solid basis for ZRT and LP motion. When driving in a limited parking place or congested streets,



Fig. 26. Road tests of the proposed space-saving 4WS4WD EV. (a)–(d) LP application in AN extremely narrow space. (e)–(h) ZRT application in a crowded parking stall.

high maneuverability and flexibility are required. The driver can control the OIS to turn the four wheels independently with the assistance of steer-by-wire technologies. Each wheel can be steered independently and driven independently depending on the steer-by-wire technology and drive-by-wire technology. The vehicle can achieve ZRT motion and LP motion via co-operation of the four steering motors and four driving motors. The maneuverability of the vehicle is greatly improved, and the challenge of operation is significantly lessened. Finally, the control strategy for the space-saving steering system of the EV is presented in the integrated 2FWS and OIS, which can easily handle the EV for high-speed driving or high-maneuverability driving.

The final experimental results demonstrate the validity of our proposed system under simulated and experimental road conditions. Based on the modeling analysis, simulation and road experiments, the results demonstrate that the steering system of the space-saving 4WS4WD EV not only exhibits good performance in terms of dynamic and steady-state

characteristics when turning but also enhances the vehicle with higher maneuverability and flexibility. The simulations and practical experiments of the mechanical motion provide valuable reference for the design and validate the feasibility of omni-directional motions. The simulation results of the driving motors indicate that the system response is rapid and smooth, and the proposed control system meets the design requirements. In the future, the proposed novel steering system of space-saving 4WS4WD EVs will be implemented on a real vehicle platform to evaluate the actual performance.

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